

Functional analysis and interpretation of RNA-seq data

June 4th, 2026

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McGill



CDSI

Computational and
Data Systems Institute



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Computational
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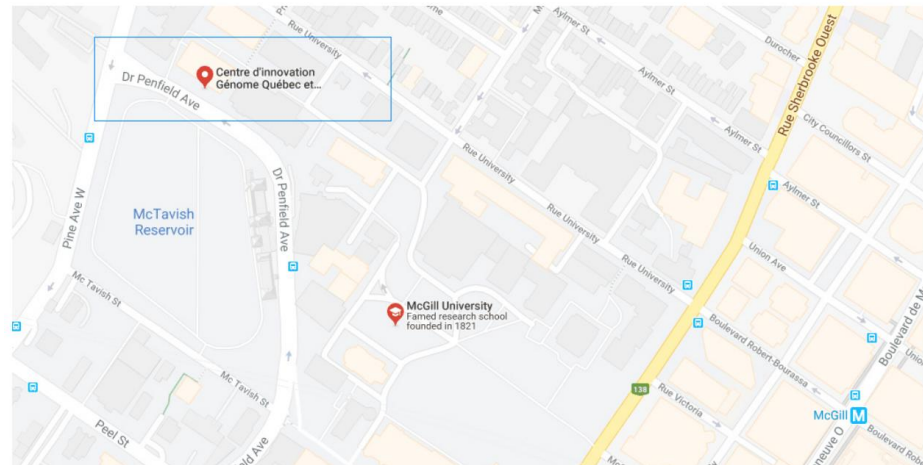


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Outline:

- 1 Experimental design and model specification using DESeq2
- 2 GSEA and ORA
- 3 Resources
- 4 Concluding remarks

Acknowledgements

Material:

Reinnier Padilla

Adrien Osakwe

Exercises:

DESeq2 vignette

fgsea vignette

Enrichr vignette

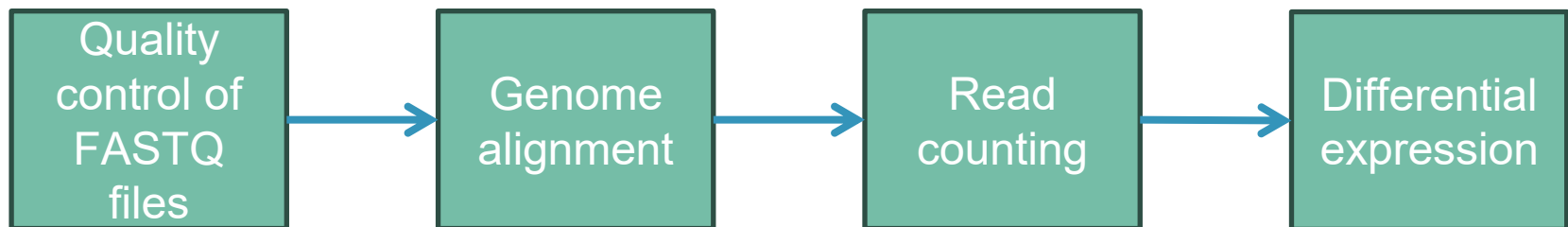
Data:

[Link](#)

This is an interactive workshop :)

Feel free to interrupt or raise your hand to ask questions

A small recap...



Part 1: Experimental design and model specification using DESeq2

- A brief overview of the DESeq2 model
- Introduction to design matrices for gene expression experiments
- Interpreting coefficients
- Multi-factor designs
- Detecting batch effects and outliers
- Hands-on activity 1

DESeq2 output

log2 fold change (MLE): ConditionCoV2

Wald test p-value: ConditionCoV2

DataFrame with 14700 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
A1BG	90.4475	-0.1439495	0.118021	-1.219695	0.222581	0.803126
A1BG-AS1	278.3007	-0.0480946	0.112076	-0.429126	0.667832	0.963881
A2M	873.3212	-0.8092580	0.741401	-1.091525	0.275042	0.833286
A2M-AS1	11.3404	0.4892287	0.256631	1.906348	0.056605	0.531853
A4GALT	775.8491	0.1133252	0.223626	0.506761	0.612322	0.951852

What does this all mean?

Love, M. *et al* (2014) *Genome Biology*

Model used in DESeq2

DESeq2 models the counts using a negative binomial distribution

$$K_{ij} \sim \text{NB}(\text{mean} = \mu_{ij}, \text{dispersion} = \alpha_i) \quad (1)$$
$$\mu_{ij} = s_{ij}q_{ij}$$

$$\log q_{ij} = \sum_r x_{jr} \beta_{ir}. \quad (2)$$

K_{ij} counts of reads for gene i , sample j

μ_{ij} fitted mean

α_i gene-specific dispersion

s_j sample-specific size factor

s_{ij} gene- and sample-specific normalization factor

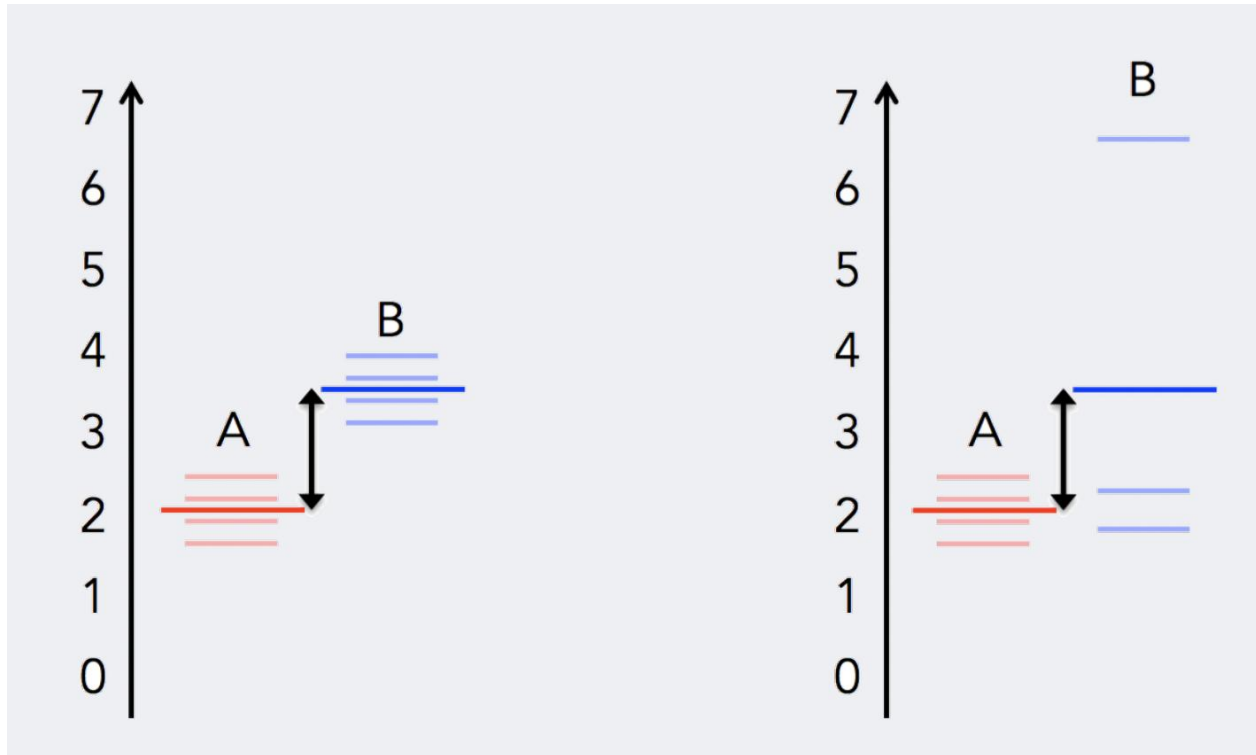
q_{ij} proportional to true concentration of fragments

x_{jr} elements of the design matrix X

β_{ir} the logarithmic fold change for gene i and covariate r

Love, M. *et al* (2014) *Genome Biology*

Uneven distribution of information



Replication introduces variance

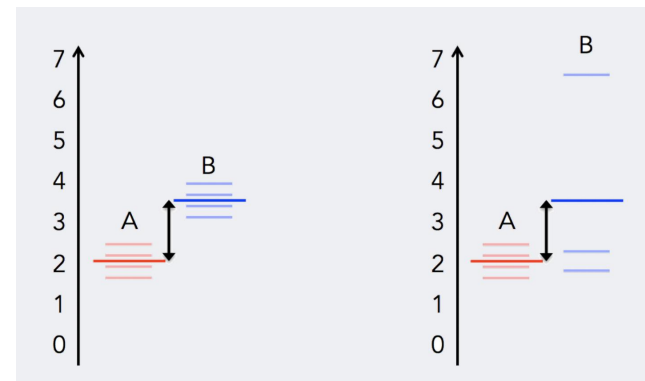
Adapted from: https://hbctraining.github.io/DGE_workshop_salmon_online/schedule/links-to-lessons.html

What is dispersion?

Within-group variability (the variability between replicates), is modeled by the dispersion parameter

$$K_{ij} \sim \text{NB}(\text{mean} = \mu_{ij}, \text{dispersion} = \alpha_i)$$

$$\text{Var } K_{ij} = \mu_{ij} + \alpha_i \mu_{ij}^2$$



K_{ij} counts of reads for gene i , sample j

μ_{ij} fitted mean

α_i gene-specific dispersion

Love, M. *et al* (2014) *Genome Biology*

Wald tests

Log2 Fold Change in gene expression

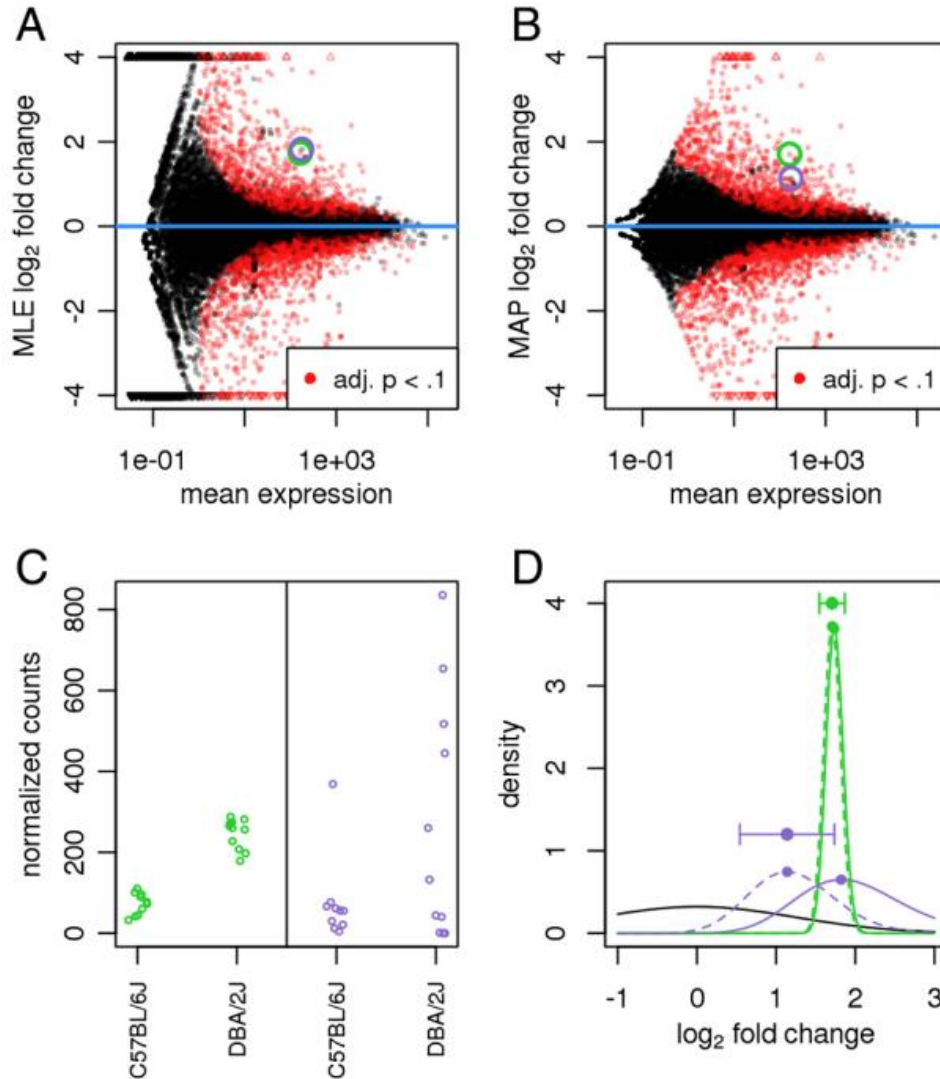
$$W = \frac{\hat{\beta}}{SE(\hat{\beta})},$$

We can assume the log fold change under the null hypothesis to be distributed normally with mean zero and variance that depends on the standard error.

From there, we can compute a two-sided p-value

Love, M. *et al* (2014) *Genome Biology*

Log-fold shrinkage



Shrinkage of FC addresses two issues:

- Log-fold changes are noisier when counts are low
- Genes with high dispersion should be less reliable

Love, M. *et al* (2014) *Genome Biology*

Part 1: Experimental design and model specification using DESeq2

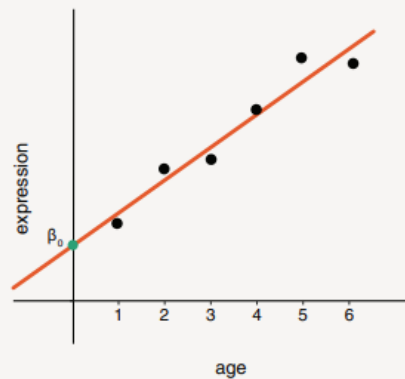
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Models for covariate and factor variables

Covariates: quantitative measurements (e.g. age)

Regression model

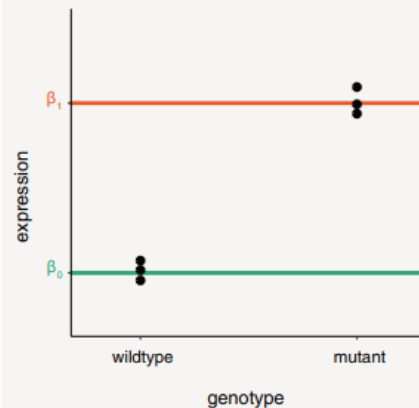
$$\text{expression} = \beta_0 + \beta_1 \text{age}$$



Factors: categorical variables (e.g. genotype)

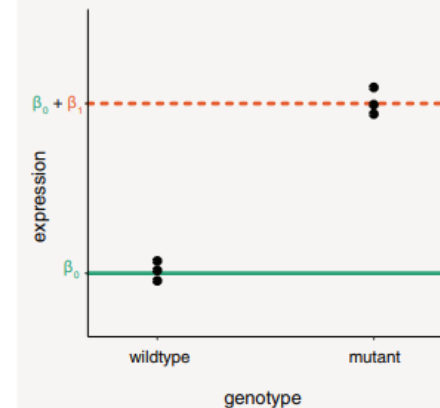
Means model

$$\text{expression} = \beta_1 \text{wildtype} + \beta_2 \text{mutant}$$



Mean-reference model

$$\text{expression} = \beta_1 + \beta_2 \text{mutant}$$



Legend

- Original data points

— Expected gene expression
(based on model)

- - - Expected gene expression
(of non-reference levels in mean-reference model)

Design and contrast matrices

Design matrix

Columns are associated with model parameters

Rows are associated with samples

$$\begin{matrix} & \begin{matrix} B1 \\ B2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \end{matrix}$$

Contrast matrix

Columns represent a contrast of interest

Rows are associated with model parameters

$$\begin{matrix} & \begin{matrix} B1 \\ B2 \end{matrix} \\ \begin{matrix} B1 \\ B2 \end{matrix} & \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \end{matrix}$$

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Gene expression modelled by a group factor

~ group

Model

$$E(y) = 2.95 + 1.62x$$

$$E(y) = 2.95$$

$$= 2.95 \quad (\text{for healthy group})$$

$$E(y) = 2.95 + 1.62$$

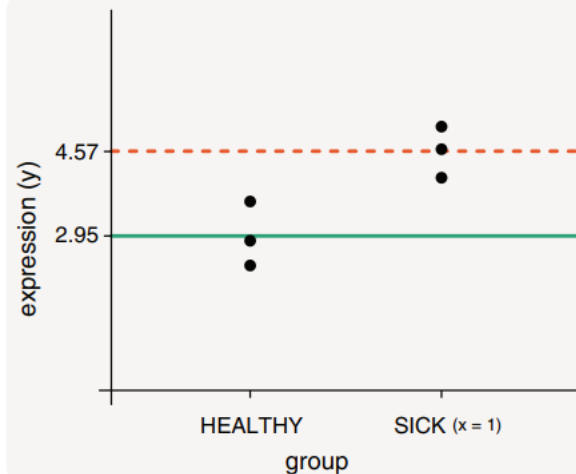
$$= 4.57 \quad (\text{for sick group})$$

Matrix

```
> model.matrix(~group)
```

$$\begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{array}{c} \text{(Intercept)} \\ \text{groupSICK} \end{array}$$

Plot



Law, C. et al (2020) F1000 Research

Gene expression modelled by a group factor (excluding intercept)

$\sim 0 + \text{group}$

A contrast will be needed to extract the actual log2FC

Model

$$E(y) = 2.95x_1 + 4.57x_2$$

$$E(y) = 2.95 = 2.95 \quad (\text{for healthy group})$$

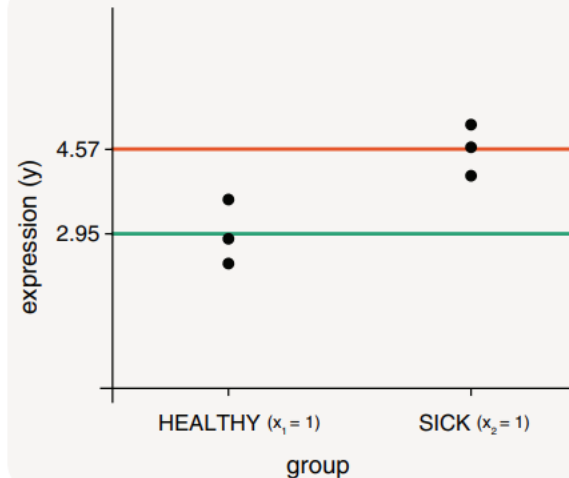
$$E(y) = 4.57 = 4.57 \quad (\text{for sick group})$$

Matrix

```
> model.matrix(~0 + group)
```

$$\begin{array}{c} \text{groupHEALTHY} \\ \text{groupSICK} \end{array} \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$$

Plot



Law, C. et al (2020) F1000 Research

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An interaction model captures a synergistic effect

$\sim \text{treat1} * \text{treat2}$ equivalent to $\sim \text{treat1} + \text{treat2} + \text{treat1}:\text{treat2}$

Model

$$E(y) = 1.03 + 1.09x_1 + 1.97x_2 + 0.82x_1x_2$$

$$E(y) = 1.03 = 1.03 \quad (\text{for control})$$

$$E(y) = 1.03 + 1.09 = 2.12 \quad (\text{for treatment I})$$

$$E(y) = 1.03 + 1.97 = 3.00 \quad (\text{for treatment II})$$

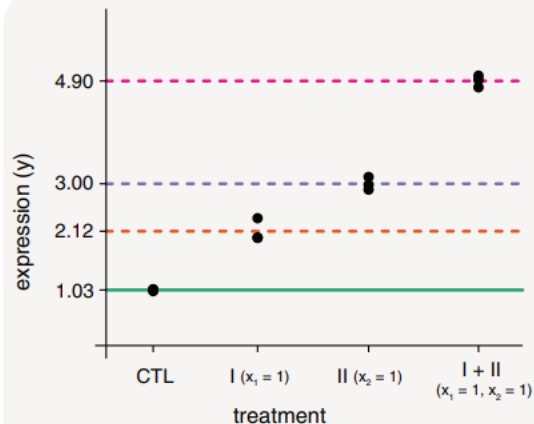
$$E(y) = 1.03 + 1.09 + 1.97 + 0.82 = 4.90 \quad (\text{for treatments I \& II})$$

Matrix

```
> model.matrix(~treat1 * treat2)
```

$$\begin{matrix} & \text{(Intercept)} & \text{treat1YES} & \text{treat2YES} & \text{treat1YES:treat2YES} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \end{matrix}$$

Plot



One plus one is greater than two

Law, C. et al (2020) F1000 Research

A group factor can represent two factors
(tissue sample and cell type)

##	expression	id	tissue	cells	group
## 1	1.01	MOUSE1	LUNG	B	LUNG_B
## 2	1.04	MOUSE2	LUNG	B	LUNG_B
## 3	1.04	MOUSE3	LUNG	B	LUNG_B
## 4	1.99	MOUSE4	BRAIN	B	BRAIN_B
## 5	2.36	MOUSE5	BRAIN	B	BRAIN_B
## 6	2.00	MOUSE6	BRAIN	B	BRAIN_B
## 7	2.89	MOUSE7	LUNG	T	LUNG_T
## 8	3.12	MOUSE8	LUNG	T	LUNG_T
## 9	2.98	MOUSE9	LUNG	T	LUNG_T
## 10	5.00	MOUSE10	BRAIN	T	BRAIN_T
## 11	4.92	MOUSE11	BRAIN	T	BRAIN_T
## 12	4.78	MOUSE12	BRAIN	T	BRAIN_T

A group factor can represent two factors (tissue sample and cell type)

$\sim 0 + \text{group}$

Model

$$E(y) = 1.03x_1 + 2.12x_2 + 3.00x_3 + 4.90x_4$$

$$E(y) = 1.03 = 1.03 \quad (\text{for lung B-cells})$$

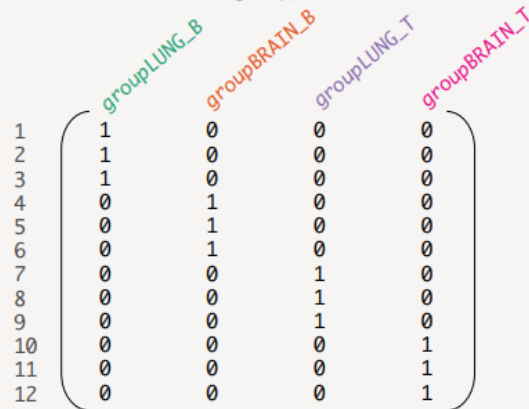
$$E(y) = 2.12 = 2.12 \quad (\text{for brain B-cells})$$

$$E(y) = 3.00 = 3.00 \quad (\text{for lung T-cells})$$

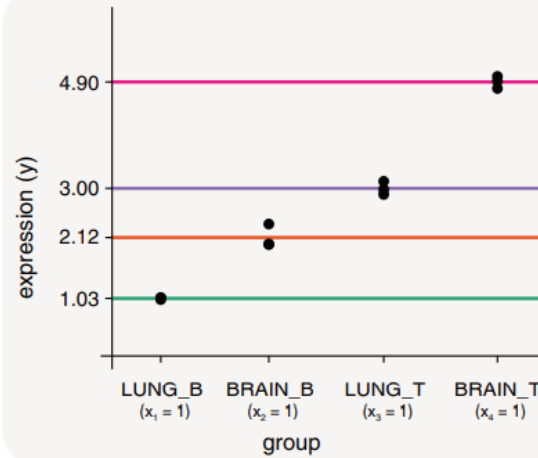
$$E(y) = 4.90 = 4.90 \quad (\text{for brain T-cells})$$

Matrix

```
> model.matrix(~0 + group)
```



Plot

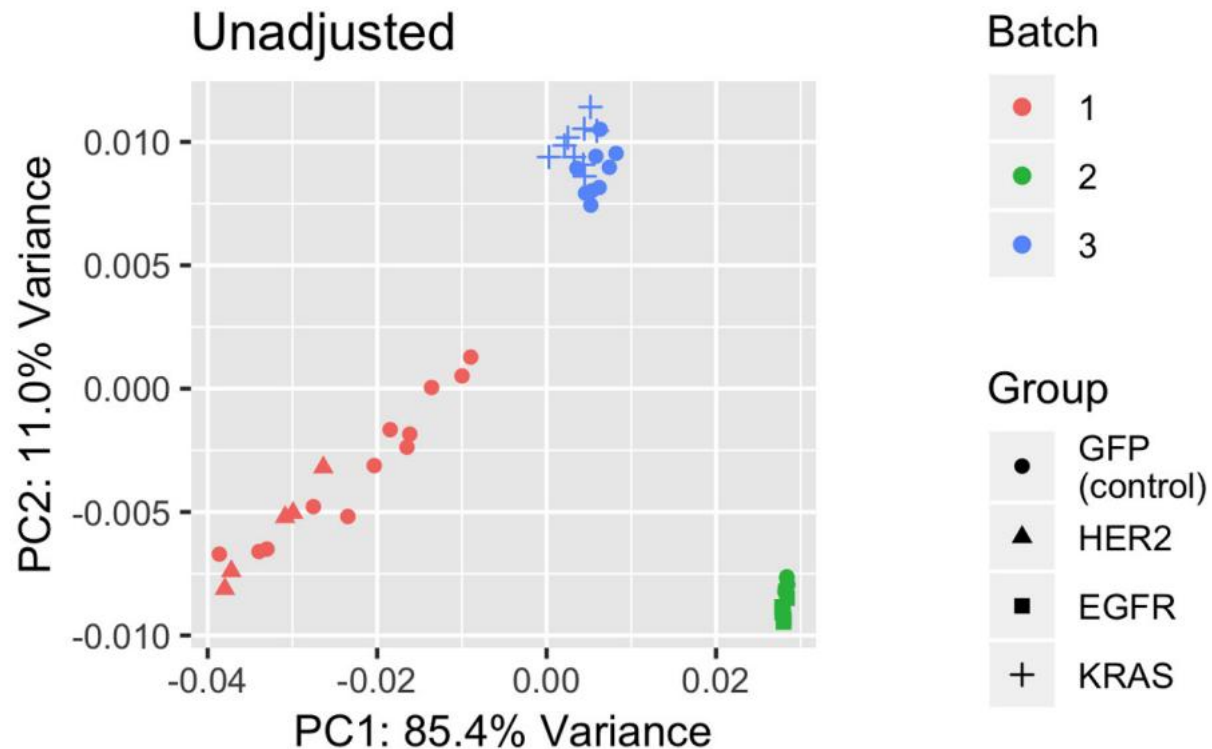


Law, C. et al (2020) F1000 Research

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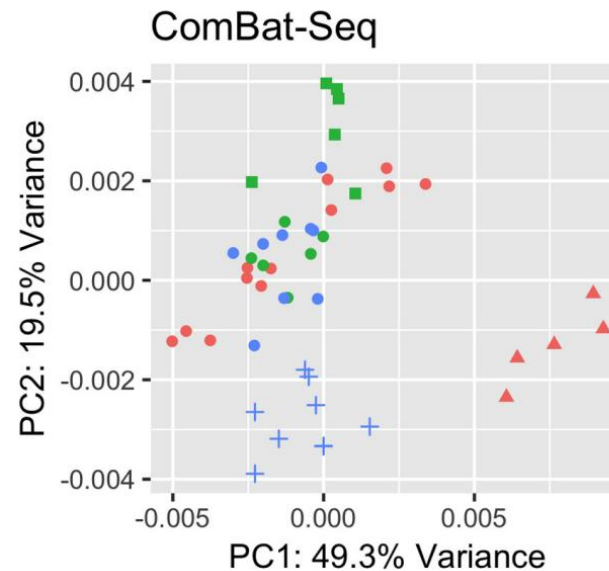
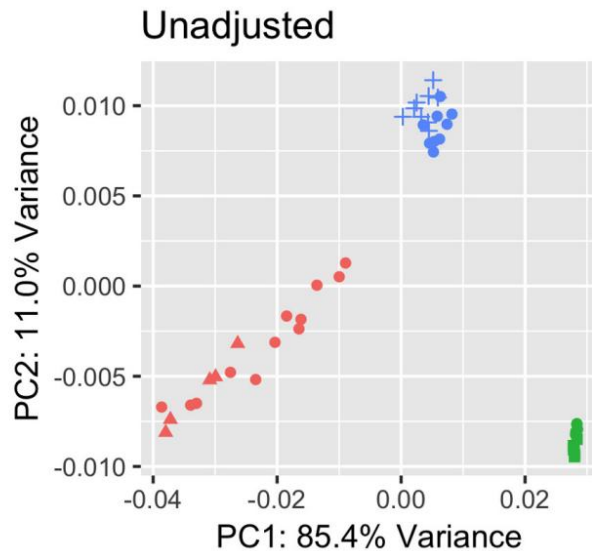
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Batch effects

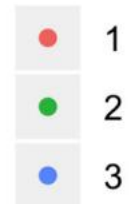


Zhang, Y. *et al* (2014) *NAR Genomics and Bioinformatics*

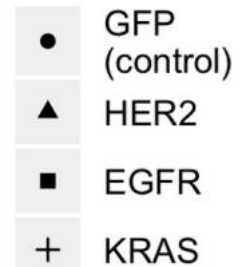
Batch effect removal



Batch



Group



Zhang, Y. *et al* (2014) *NAR Genomics and Bioinformatics*

DESeq2 output

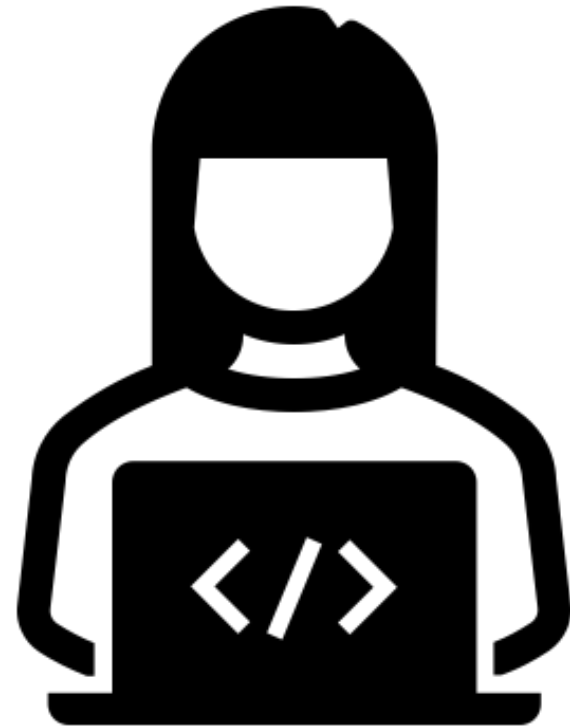
log2 fold change (MLE): ConditionCoV2
Wald test p-value: ConditionCoV2
DataFrame with 14700 rows and 6 columns

The log2 fold change depends on the design that we use and the contrast (if any)

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
A1BG	90.4475	-0.1439495	0.118021	-1.219695	0.222581	0.803126
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Love, M. *et al* (2014) *Genome Biology*

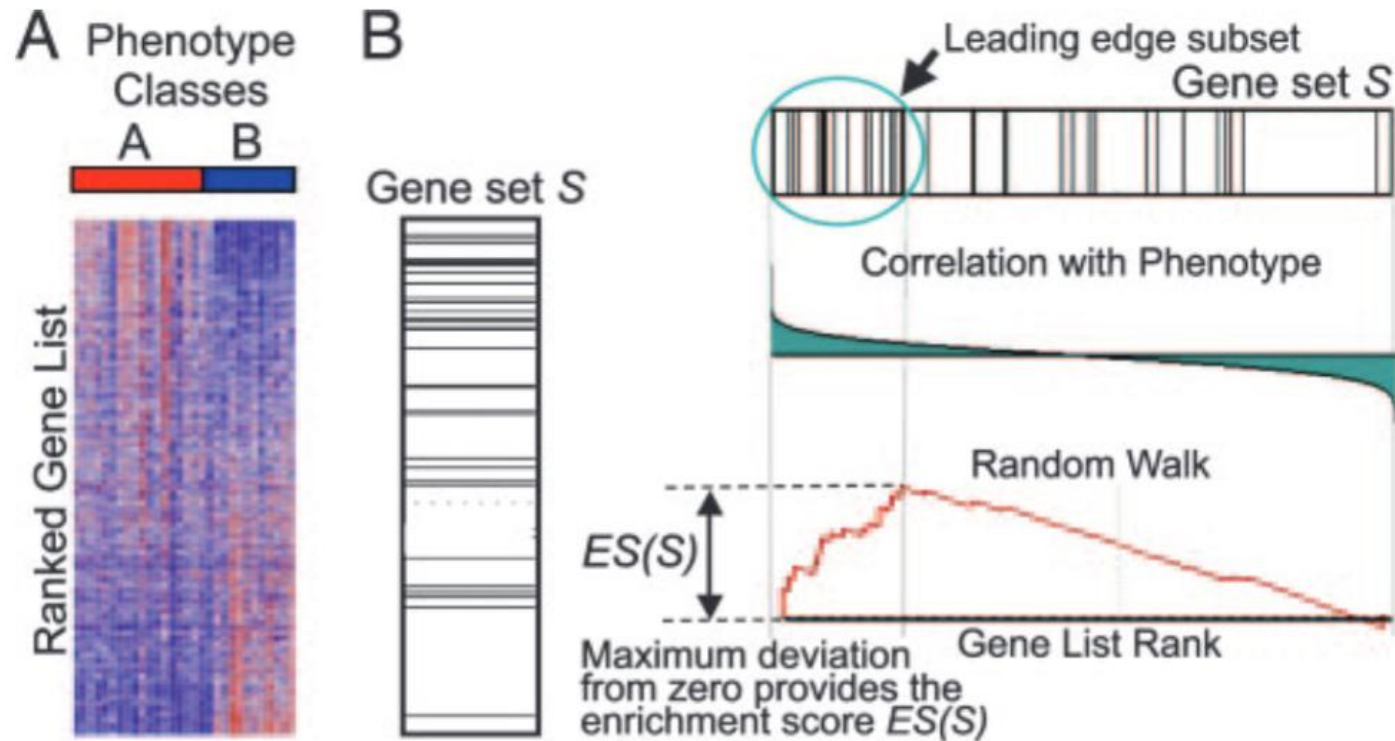
Hands-on 1



Part 2: GSEA and ORA

- Statistical concepts used in GSEA
- Statistical concepts used in ORA
- When to use GSEA or ORA?
- Hands-on activity 2

Gene Set Enrichment Analysis (GSEA)



Subramanian, A. *et al* (2005) *PNAS*

Part 2: GSEA and ORA

- Statistical concepts used in GSEA
- Statistical concepts used in ORA
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- Hands-on activity 2

Enrichr



Enrichr

	In Pathway	Not in Pathway	Total
Selected genes	k	$n - k$	n
Not selected genes	$K - k$	$N - K - (n - k)$	$N - n$
Total	K	$N - K$	N

Where:

- N = total number of genes in the background (e.g. all detected genes).
- K = number of genes in the pathway.
- n = number of genes selected as "significant" (e.g. DE genes).
- k = overlap between selected genes and pathway genes.

P-value comes from a hypergeometric test

$$p = \sum_{i=k}^{\min(n,K)} \frac{\binom{K}{i} \binom{N-K}{n-i}}{\binom{N}{n}}$$

Chen E. *et al* (2013) *BMC Bioinformatics*

Part 2: GSEA and ORA

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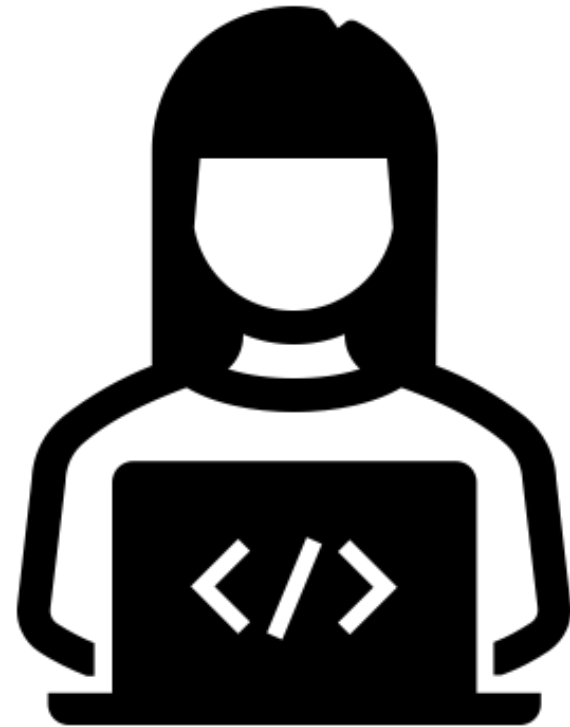
When to use GSEA or ORA?

	GSEA	ORA
Input	Ranked list of all genes	Significant gene list + background
Test type	Running-sum enrichment score + permutation	Hypergeometric
Uses cutoff?	No	Yes(DE)

ORA asks: Are my DE genes enriched in this pathway?

GSEA asks: Are genes from this pathway enriched towards the top or bottom of the ranked list?

Hands-on 2



Part 3: Resources

MsigDB

Gene Ontology

KEGG

Reactome

Enrichr

Panther



Part 4: Concluding remarks

Now you are ready to:

- Perform a standard **DGE**, **GSEA** and **ORA** analyses
- Modify design matrices to address biological questions
- Choose between GSEA and ORA depending on the question at hand

Thanks for your attention!



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References

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- Subramanian, A., Kuehn, H., Gould, J., Tamayo, P., & Mesirov, J. P. (2007). GSEA-P: a desktop application for Gene Set Enrichment Analysis. *Bioinformatics*, 23(23), 3251-3253.
- Chen, E. Y., Tan, C. M., Kou, Y., Duan, Q., Wang, Z., Meirelles, G. V., ... & Ma'ayan, A. (2013). Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool. *BMC bioinformatics*, 14(1), 128.
- Law, C. W., Zeglinski, K., Dong, X., Alhamdoosh, M., Smyth, G. K., & Ritchie, M. E. (2020). A guide to creating design matrices for gene expression experiments. *F1000Research*, 9, 1444.

Normalization in DESeq2

Median of ratios

$$\mu_{ij} = s_{ij} q_{ij}$$

Size factor

$$s_j = \text{median}_{i: K_i^R \neq 0} \frac{K_{ij}}{K_i^R} \quad \text{with}$$

$$K_i^R = \left(\prod_{j=1}^m K_{ij} \right)^{1/m} .$$

Geometric mean

K_{ij} counts of reads for gene i , sample j

μ_{ij} fitted mean

α_i gene-specific dispersion

s_j sample-specific size factor

s_{ij} gene- and sample-specific normalization factor

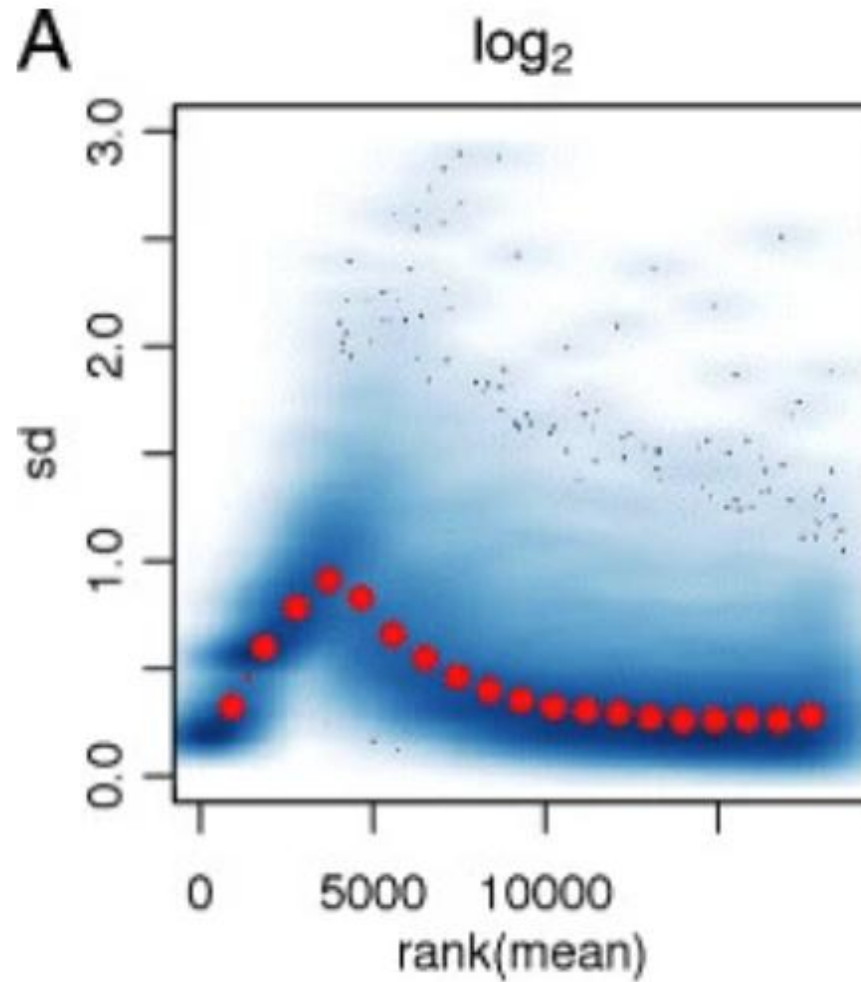
q_{ij} proportional to true concentration of fragments

Love, M. *et al* (2014) *Genome Biology*

Statistical modelling concepts used in DESeq2 and its differences with respect to edgeR

	edgeR	DESeq2
Dispersion	Negative Binomial dispersions capture global trend, quasi-dispersions account for gene-specific variability.	Gene-specific dispersion shrunk towards the trend using a MAP
Normalization	TMM	Median of ratios
Test	Quasi-likelihood F-tests	Wald-tests

Uneven distribution of information



Love, M. *et al* (2014) *Genome Biology*

Shrinkage estimation of dispersion

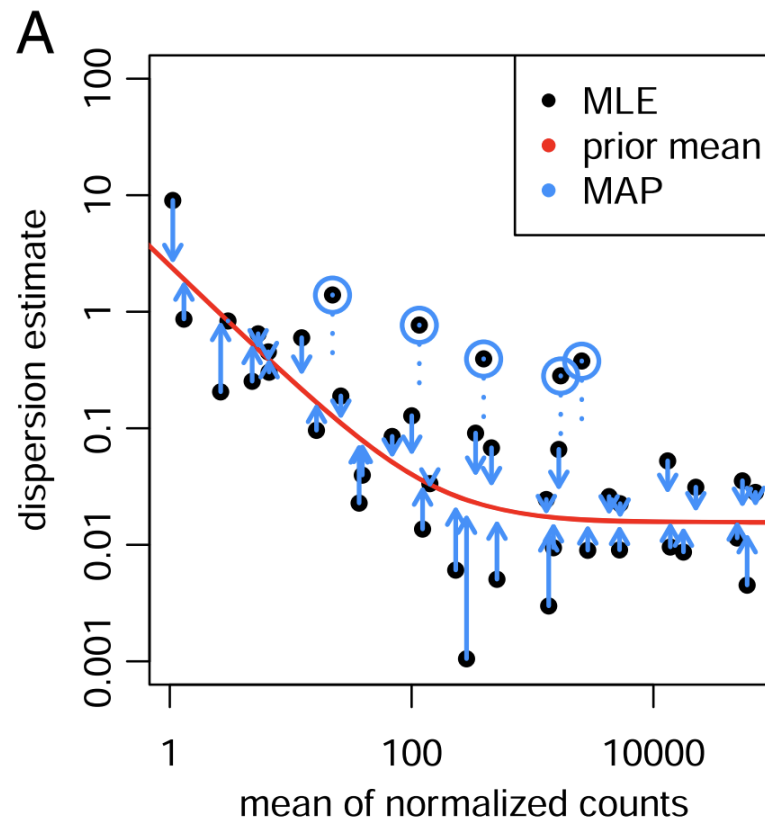


Fig 1A, adapted from: Love, M. *et al* (2014) *Genome Biology*